

1) $f(x) = \exp(x^2) + x^{-3} - 70$

$f'(x) = 2x \cdot \exp(x^2) + (-3x^{-4})$

$x_0 = 2$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

1: $2 - \frac{44.723}{218.105} = 1.795$

2: $1.795 - \frac{15.252}{89.744} = 1.625$

3: $1.625 - \frac{4.255}{44.147} = 1.529$

4: $1.529 - \frac{0.639}{37.728} = 1.508$

$x_0 = 0.5$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}$

1: $0.5 - \frac{(-0.776)}{(-46.716)} = 0.485$

2: $0.485 - \frac{0.037}{(-46.716)} = 0.486$

3: $0.486 - \frac{(-0.022)}{(-46.716)} = 0.486$

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$x_0 = -7 \quad x_1 = -7.2$

$x_{n+1} = x_n - f(x_n) \cdot \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$

1: $-7.2 - (-6.35760) \cdot \frac{(-7.2) - (-7)}{(-6.35760) - (-8.282)} = -7.86094$

2: $-7.86094 - (21.76052) \cdot \frac{(-7.86094) - (-7.2)}{21.76052 - (-6.35760)} = -7.34943$

3: $-7.34943 - (-4.22976) \cdot \frac{(-7.34943) - (-7.86094)}{(-4.22976) - 21.58745} = -7.43266$

4: $-7.43266 - (-2.55267) \cdot \frac{(-7.43266) - (-7.34943)}{(-2.55267) - (-4.22976)} = -7.42857$